

Kyligence

Big Data Acceleration Platform

Kyligence Enterprise
Kyligence Cloud

#Big Data #Enterprise Platform #AI Model Management #Paas

PROJECT INTRO

KYLIGENCE CLOUD

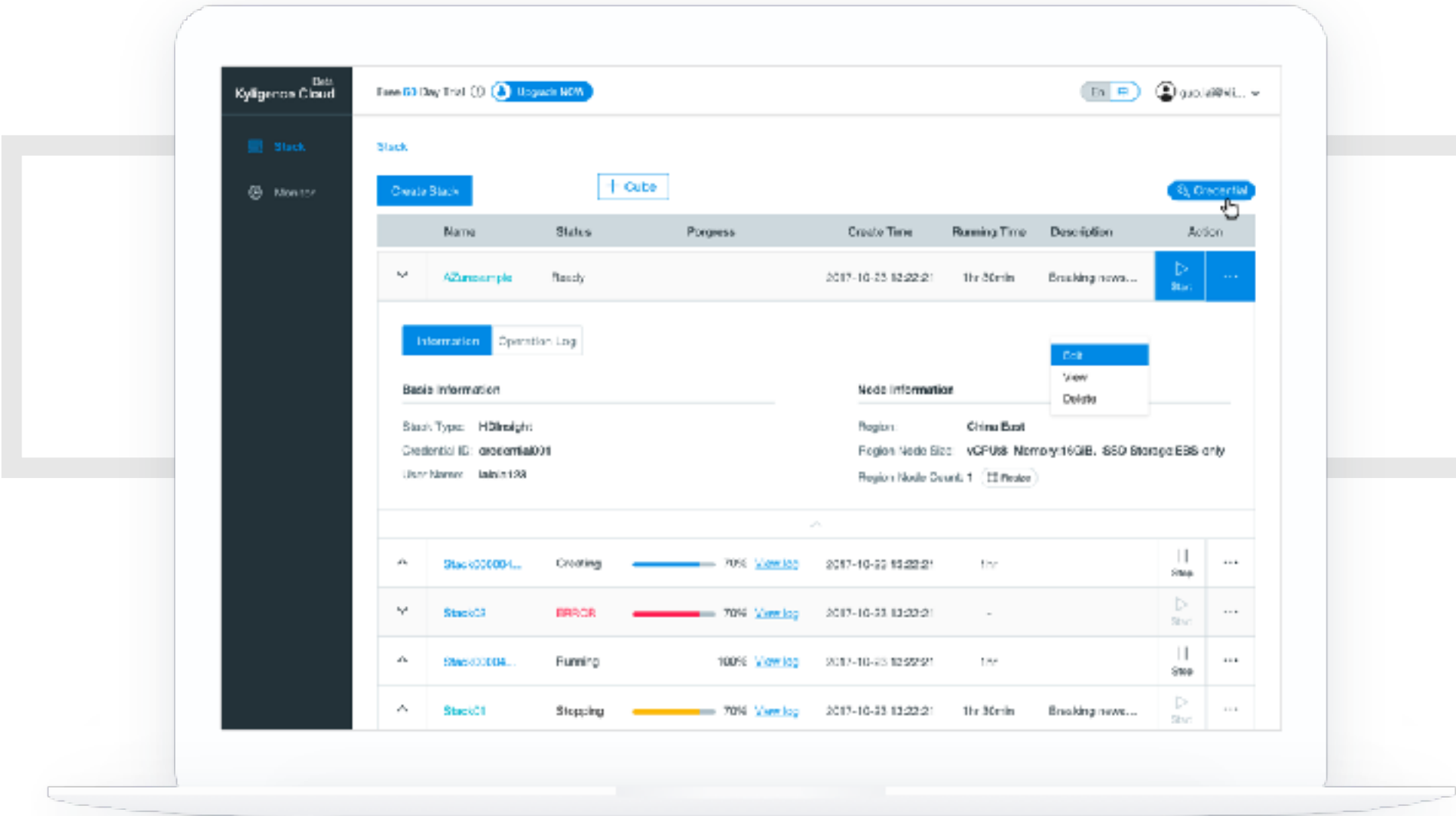
Kyligence Cloud (KC) is a IAAS platform, aim to help engineer to deploy and manage cluster on cloud. KE also can be assigned to deploy on one of cluster. When I joined Kyligence, KC was a new product need to create new UX/UI for it.

KYLIGENCE ENTERPRISE

Kyligence Enterprise (KE) is a PAAS platform, aim to help data analyst to query accelerated model by SQL and transfer pre-calculated result to BI software.

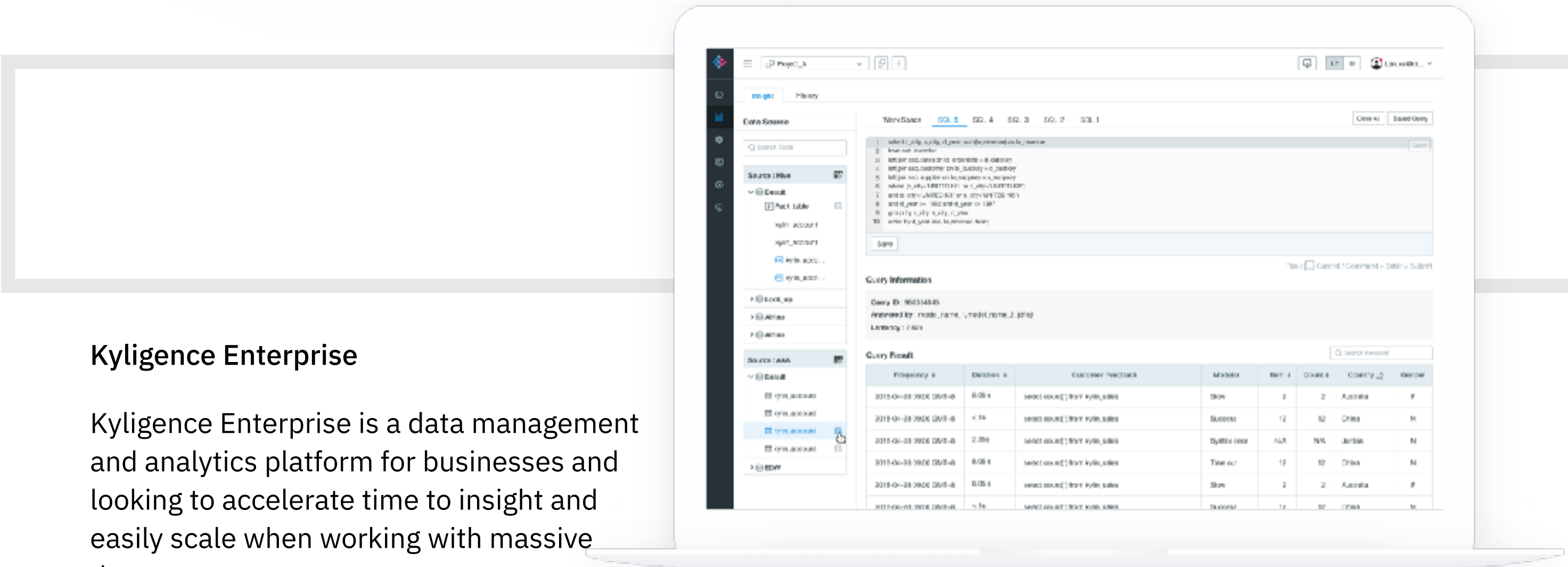
RESPONSIBILITIES

New Product Development
Design Guideline System
Prototype & UX/UI Design
Internal User Research



Kyligence Cloud

Kyligence Cloud is a big data platform offering cluster management and deployment services in the cloud.

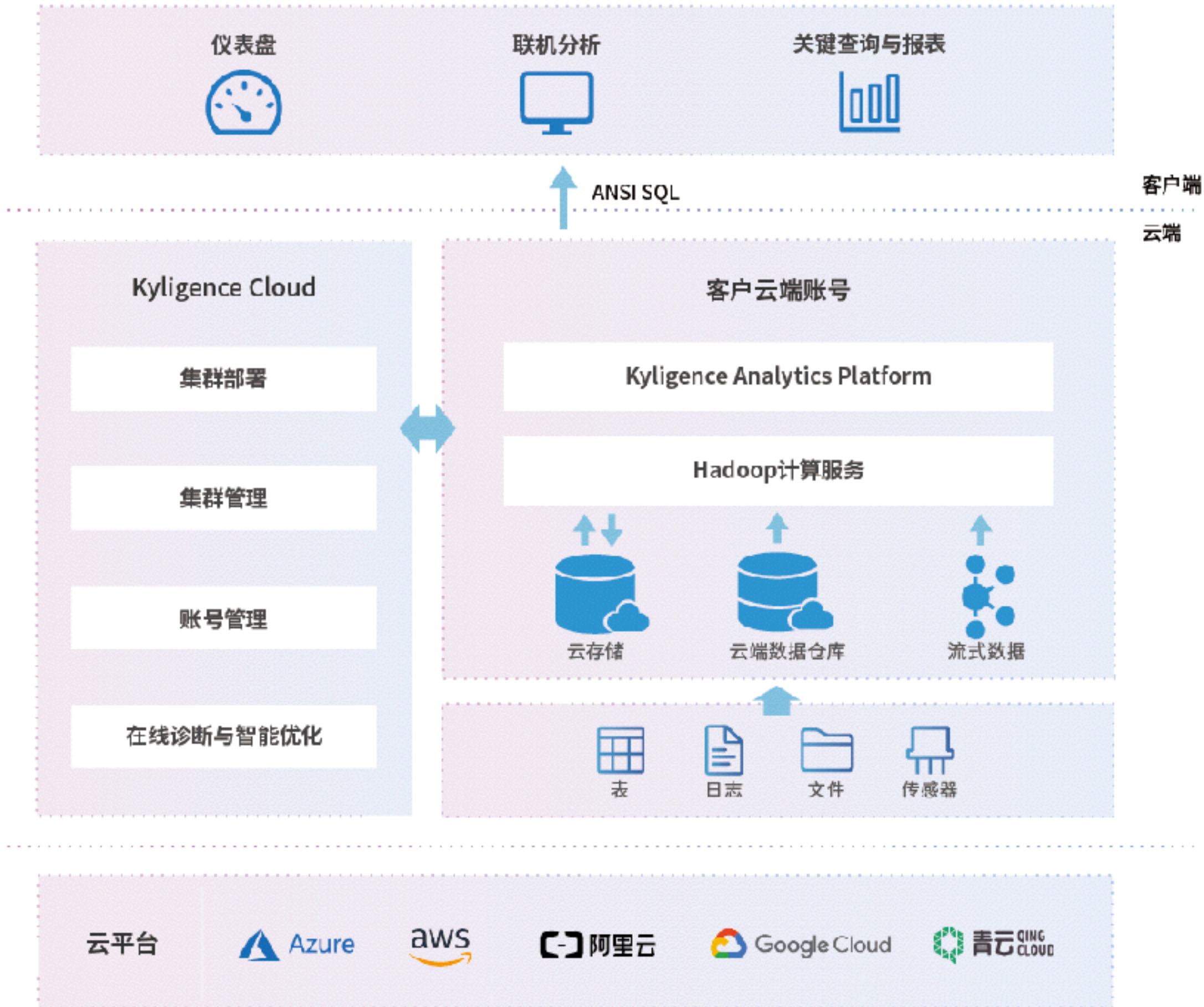




ARCHITECTURE DIAGRAM

Our clients can build up the master-slave structure on cloud by using Kyligence Cloud, deploy many clusters and assign which one is master cluster, other clusters are slave cluster. The ability of beta **Kyligence Cloud** can be allowed to connect AWS and AZURE public cloud.

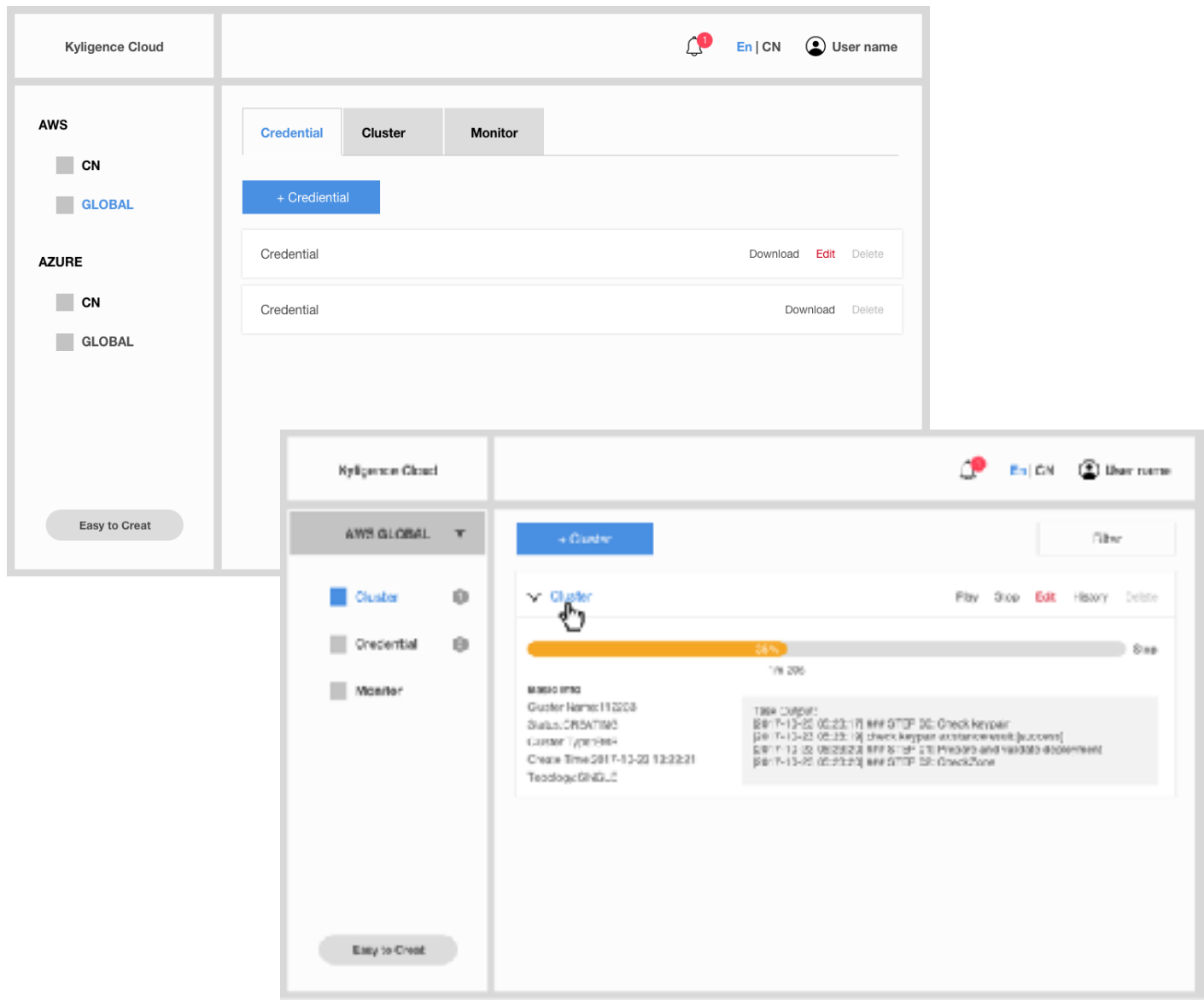
Kyligence Enterprise can be build up on one of these clusters and accelerate massive data (pre-calculating tech) that user query by SQL. At the meantime, KE can set up dev-Ops tasks at specific time to input updated data for pre-calculating model. So if our clients use SQL to query the result, BI software can see the latest diagrams transmitted by KE. As long as KE have connected to BI software, like Power BI, Tableau, Smart BI, etc.



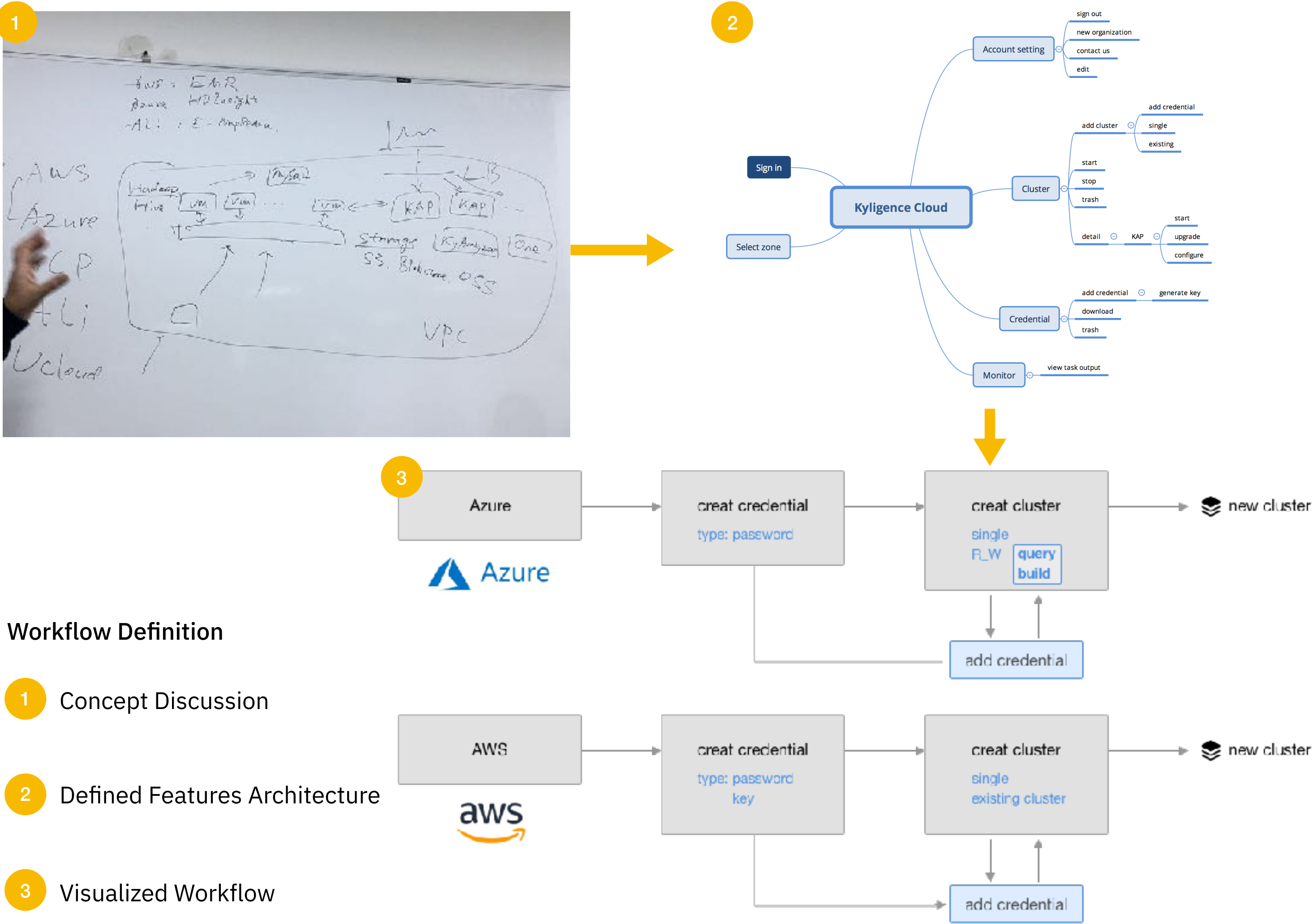
KC & KE Relational Architecture Diagram

KYLIGENCE CLOUD

KC is a IAAS platform can be deploy on public clouds or private clouds, like Azure, AWS and Ali Cloud. Because KC is a new product in Kyligence when I was joined. So we need to design new UX/UI and workflow for KC.



Output : KC Prototypes

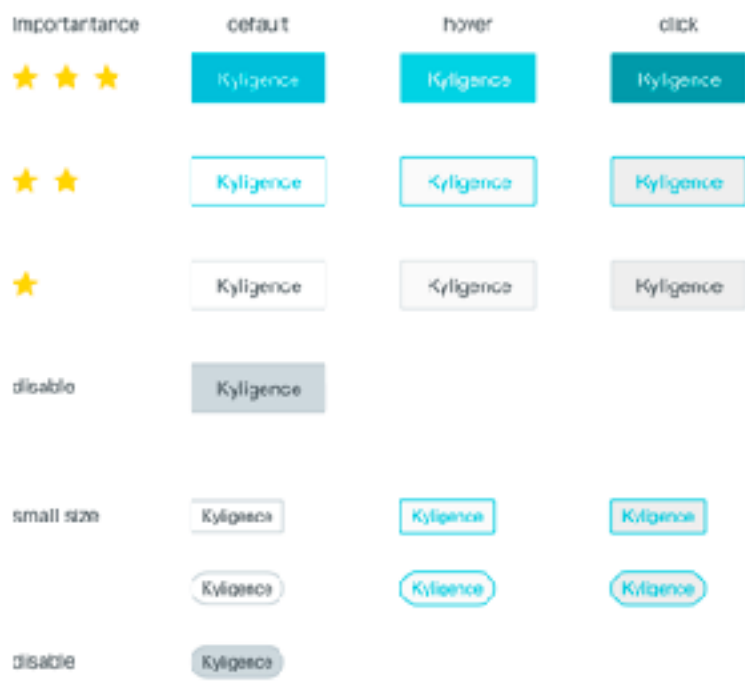


APPLY SIMPLIFIED VERSION OF GUIDELINE

Basic Setting



Color Definition

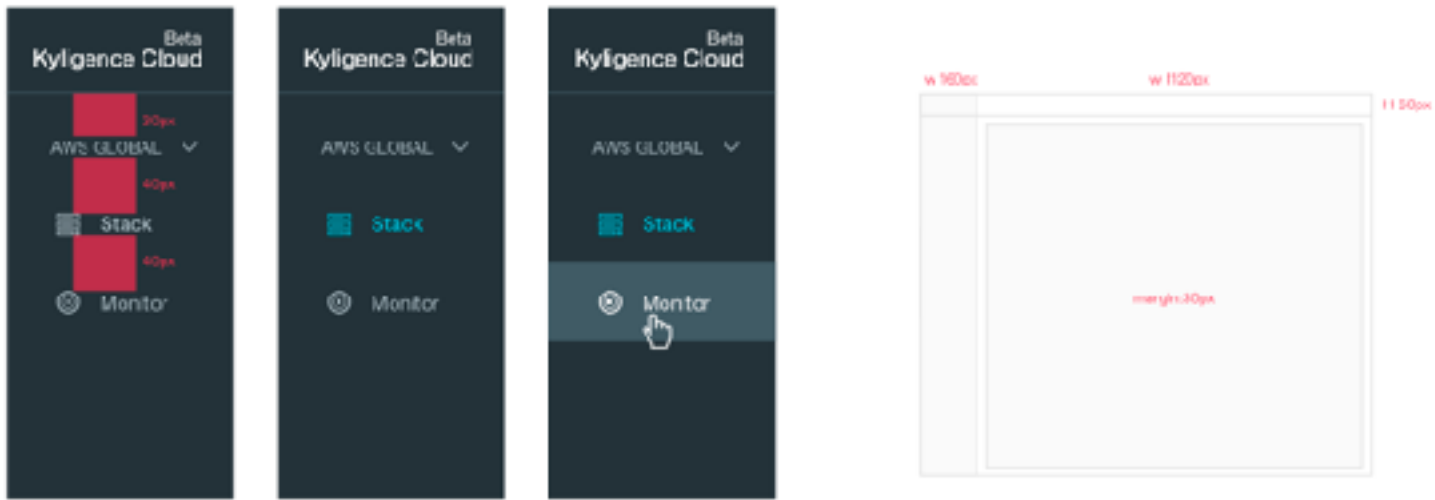


Button Definition

Component Setting

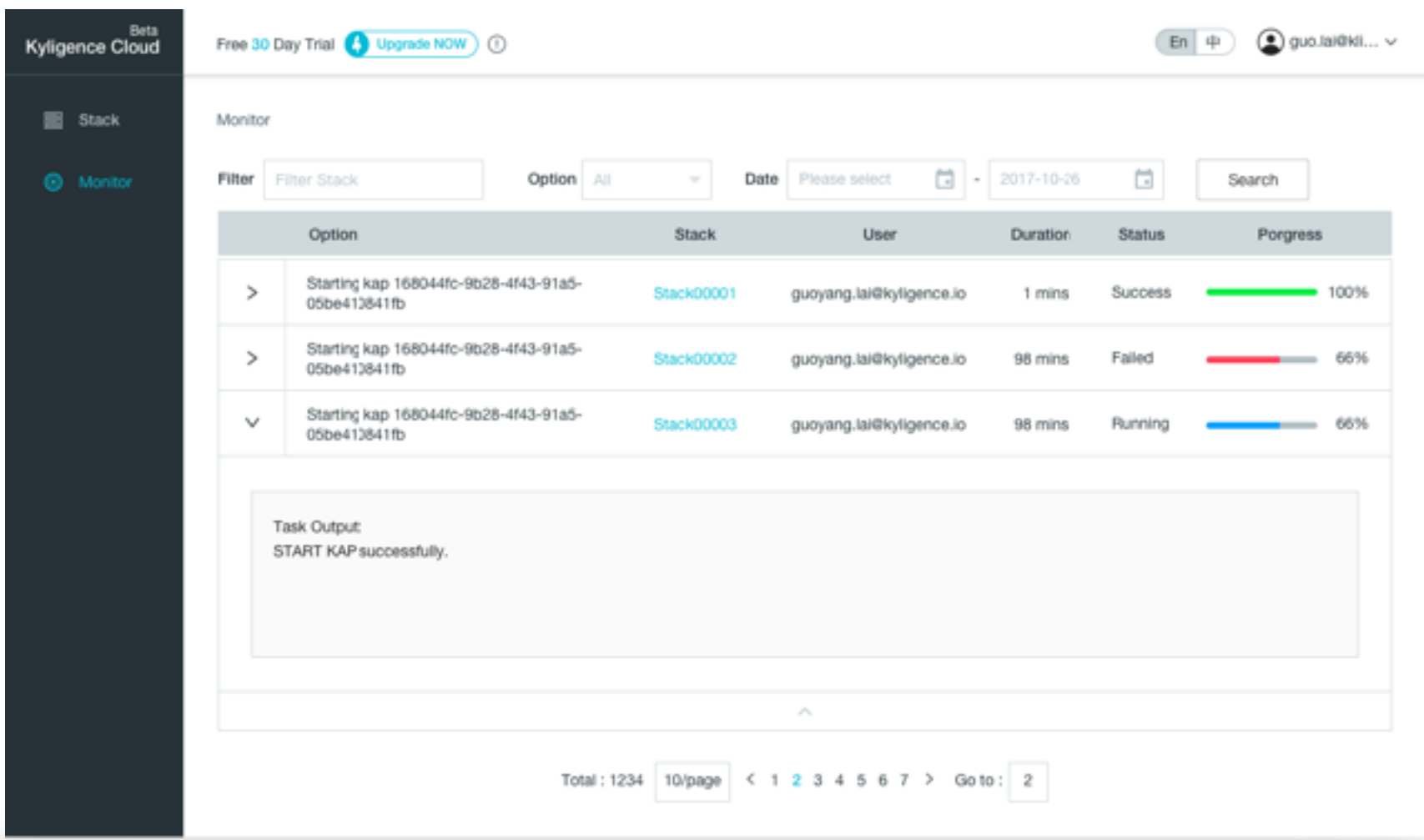
	Name	Status	Create Time	Running Time	Description	Action
Default	Stack000001...	Creating 70% View log	2017-10-23 13:22:21	1hr		Step ...
hover	Stack000001...	Creating 70% View log	2017-10-23 13:22:21	1hr		Step ...
hover	Stack000004...	Creating 70% View log	2017-10-23 13:22:21	1hr	Creeking news...	Step ...
	AZ_resample	New	2017-10-23 13:22:21	1hr 00-min		▶ Start
						⌵ Edit View Delete

Table Definition



Navigation Definition

Page



Monitor Page

Simplified Version of Guideline

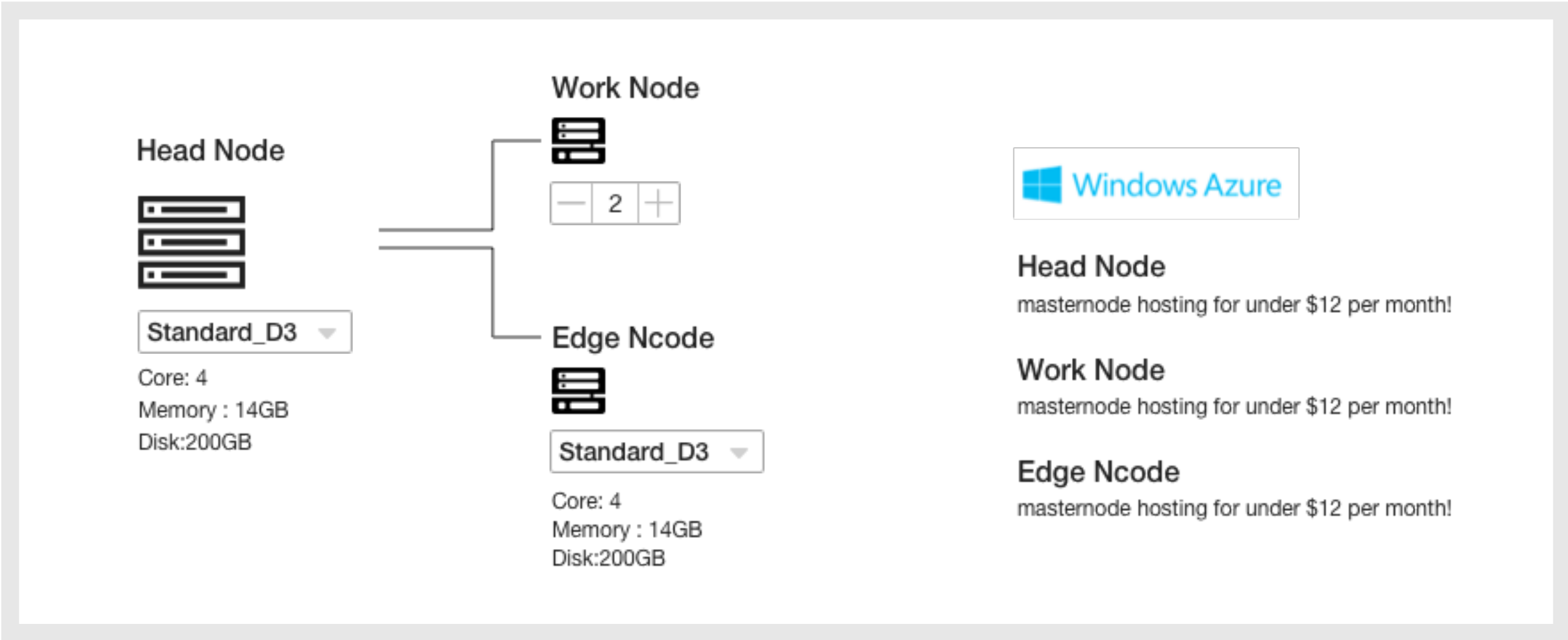
KC is a whole new product when I joined, so we decided to have beta version first before GA. Almost cost one and half weeks to complete this simplified version of guideline.

KEY FEATURE

After conceptual discussion, PM with me decide to visualize the framework of nodes management, include adding animation to remind our user how many nodes they already added, control the number of virtual physical hardware on cloud and correspond to different clouds have specific setting. So we started from drawing prototypes to have internal test, asked experts whether they could understand and operate. After several rounds of testing, we had final version and ready to GA.

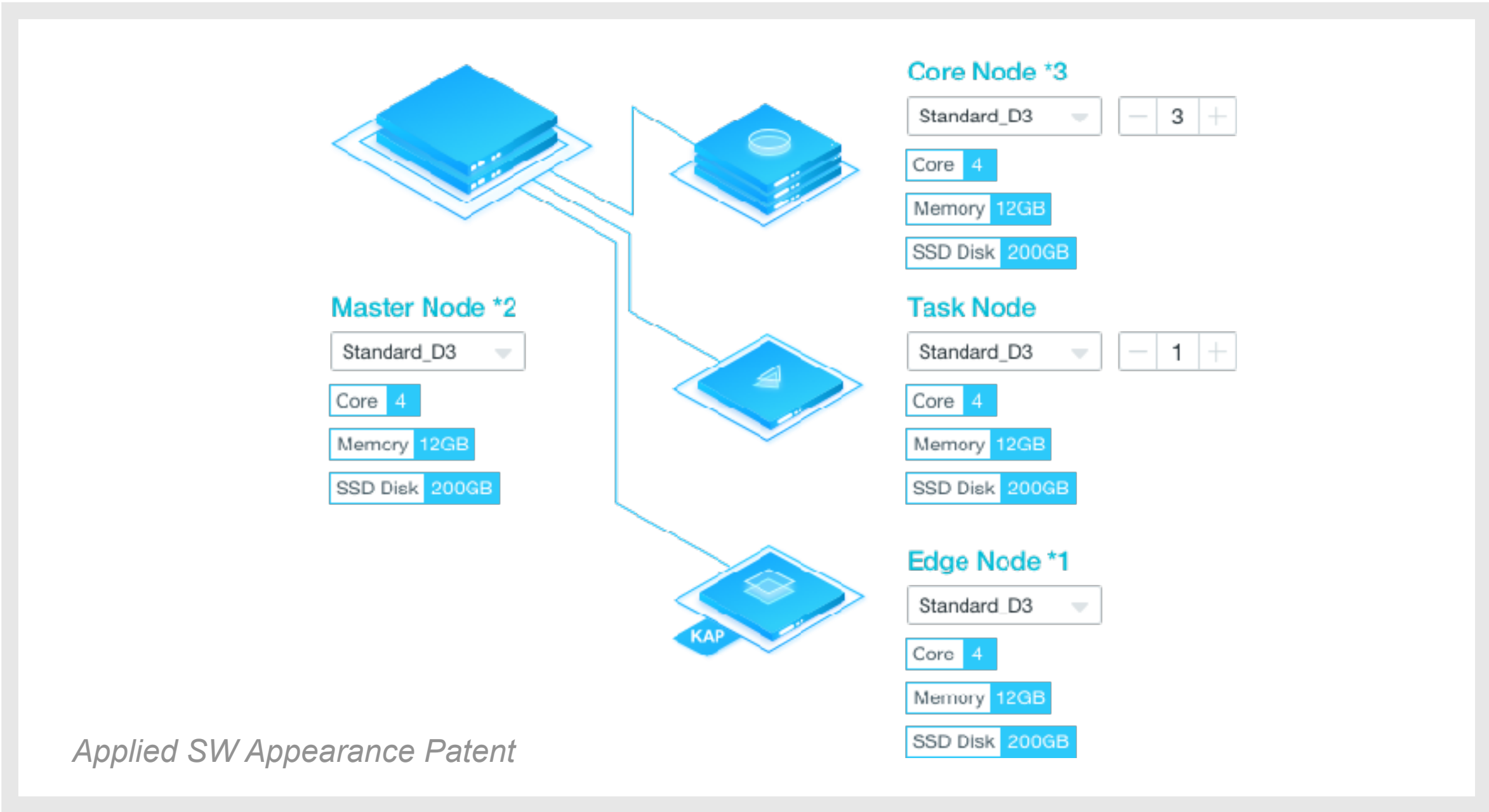
Duty of Each Node

- 1.Master Node : only responsible for commanding the duty of node.
- 2.Core Node : responsible for cloud computing execution.
- 3.Task Node : responsible for executing task commands from master.
- 4.Edge Node : only responsible for installing the KAP(KE’s old name) platform.



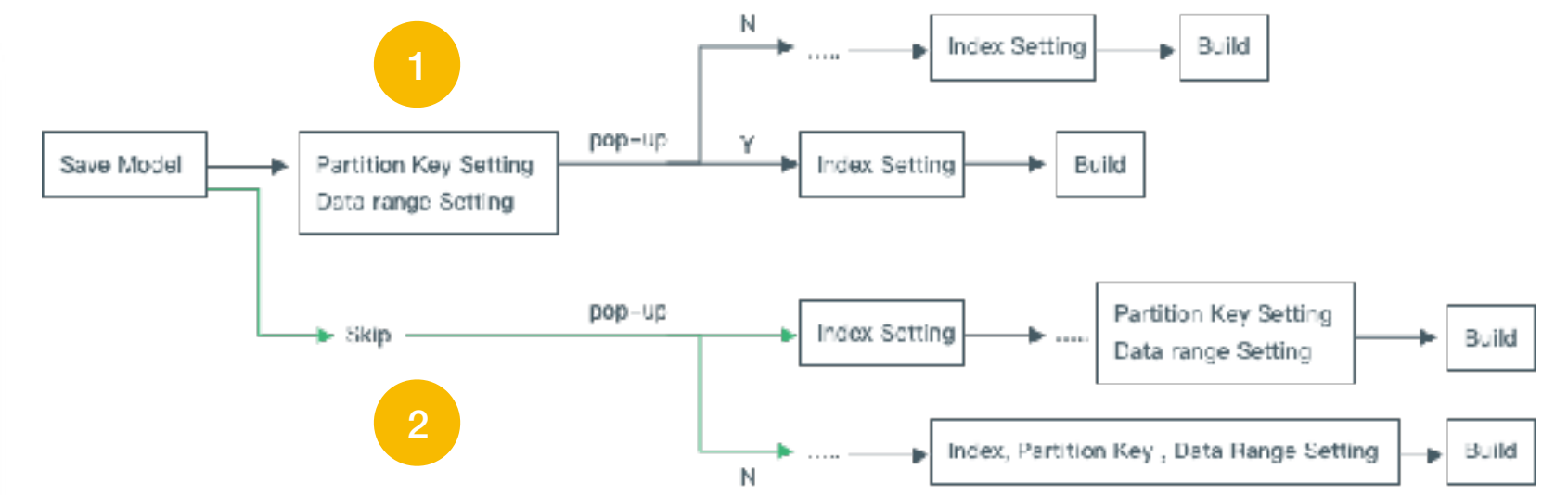
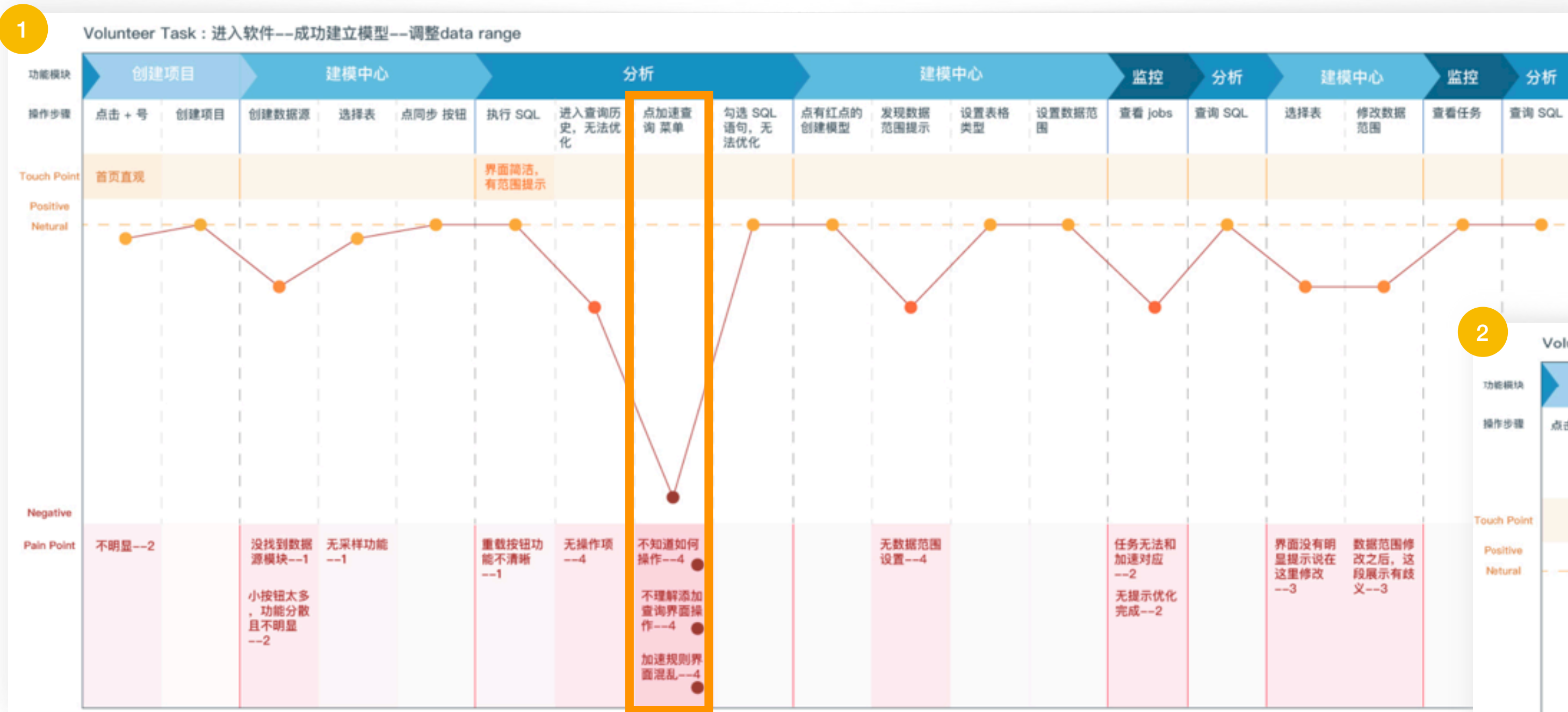
Concept of Nodes Management

Final Version of Nodes Management



Questionnaires List

User Journey Map : Task Testing



Pain Points on KE

- 1. Manual is too hard understand.
- 2. It's difficult to build up the accelerating cube.
- 3. Unfriendly with beginner.
- 4. Spend too much time to maintain the cubes.

The Most Unfriendly Feature : Accelerating Operation

- 1. Don't know how to complete the task of building cube.
- 2. Can not realize the purpose on search page.
- 3. Unable to understand the accelerating rules.



SIMPLIFY SETTING & OPERATION

In addition to a large number of steps in each process, too many redundant functions are displayed on the interface, resulting of interruption of user operation. We decided to hide rarely used buttons and add feature of batch processing , it is also necessary to strengthen reminders and operation feedbacks.

Hide Rarely Used Buttons

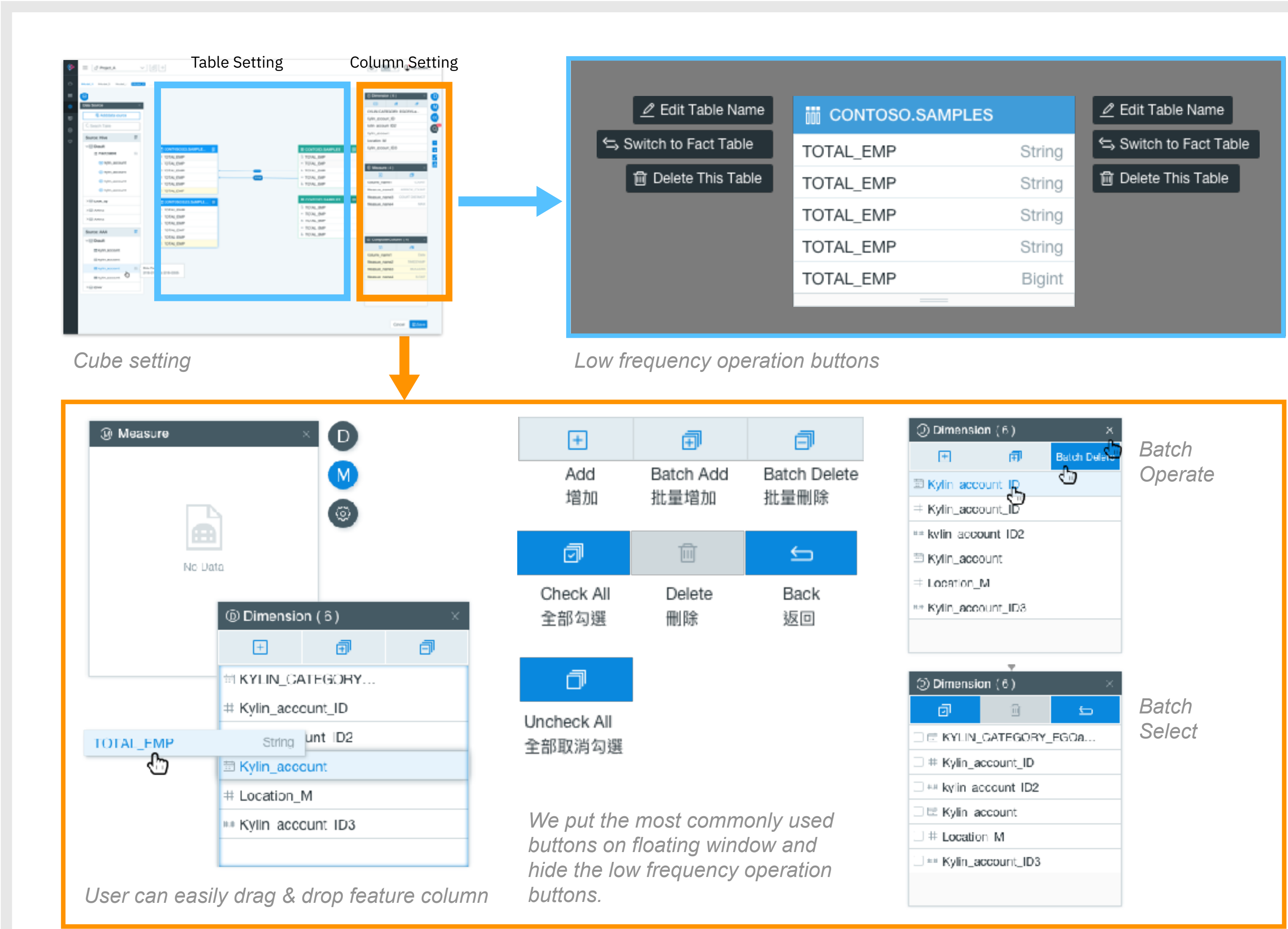
Imitated the UX of Trello to hide rarely used buttons, try to simplify the user interface.

Add Drag & Drop Feature

To help users more easily edit and drag the columns between tables.

Add Batch Processing Feature

After simplified UI, we added batch processing feature to help users save a lot of operation steps(select & delete) and time.



STRENGTHEN TUTORIAL

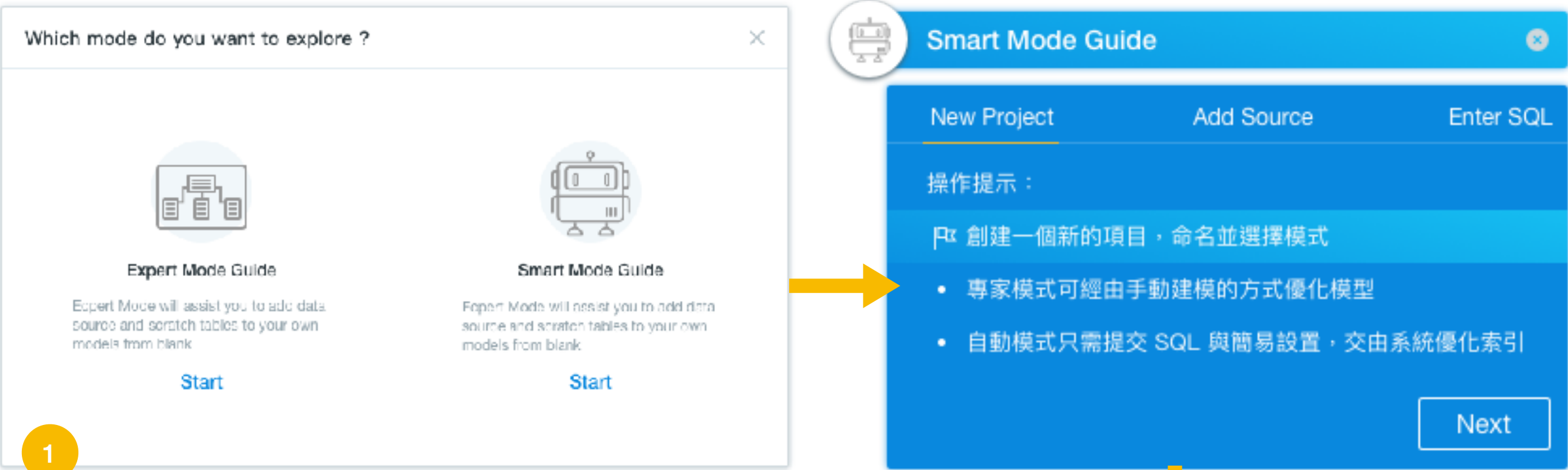
We created a new tutorial for our users to demonstrate whole process of project and cube created. The KE tutorial contain two modes both with “Expert Mode” and “Smart Mode”.

Expert Mode

Expert mode is fully demonstrate the features of KE, especially for analyst who can create semantic layer via dragging tools and query index of cube by SQL.

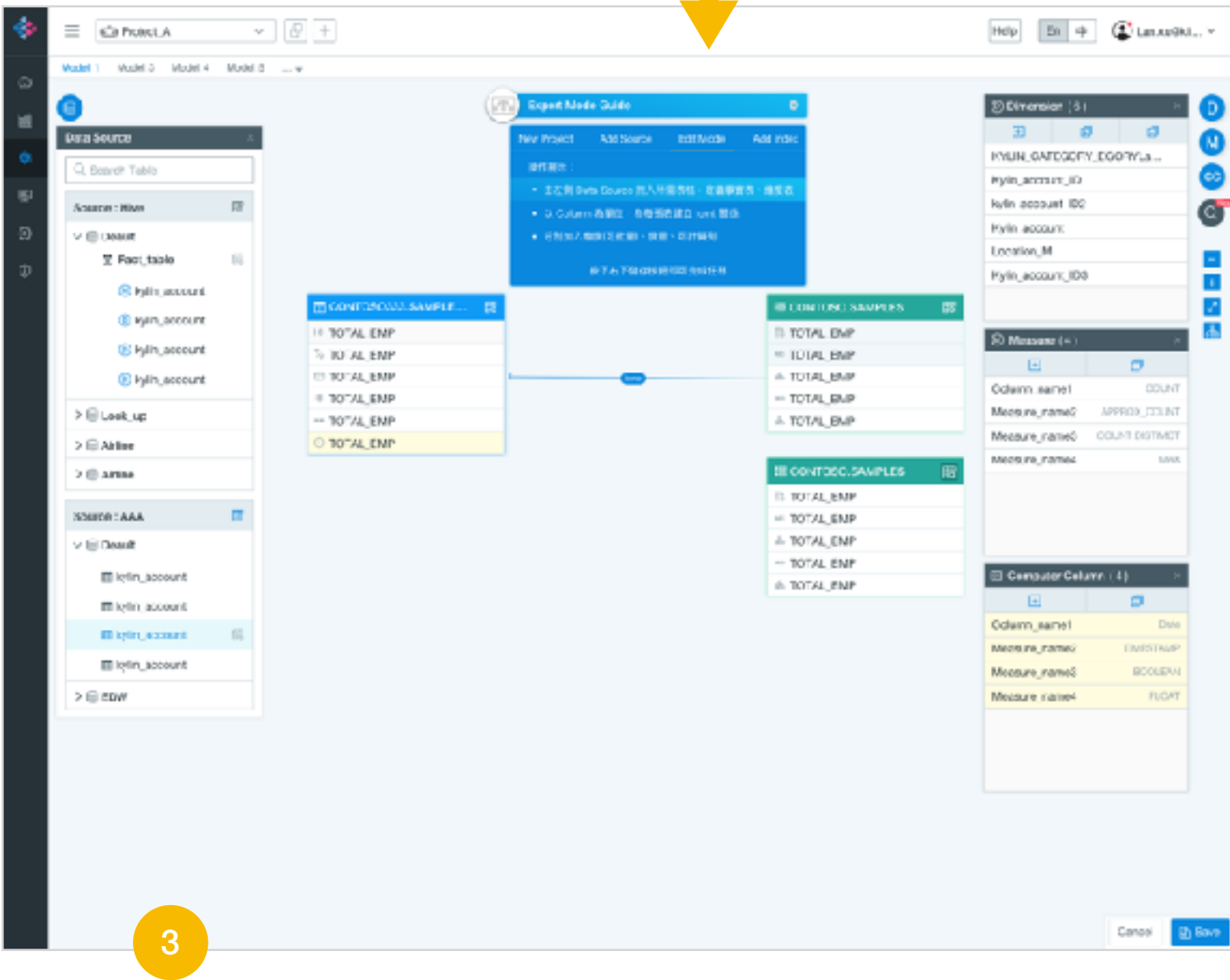
Smart Mode (for Beginner)

Smart mode is only for beginner who only can use SQL to search. KE AI engine will automatically build up the cubes by querying SQL.



TUTORIAL

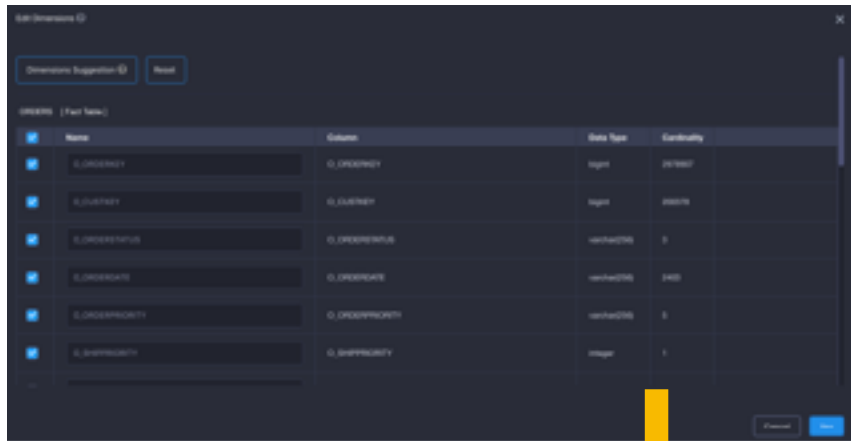
- 1 User can select one of two modes when user first enter KE, also can trigger from help list.
- 2 User can view the demo project by controlling the steps of guide .
- 3 The pop-up window can be drugged around that avoid blocking the view of the user.



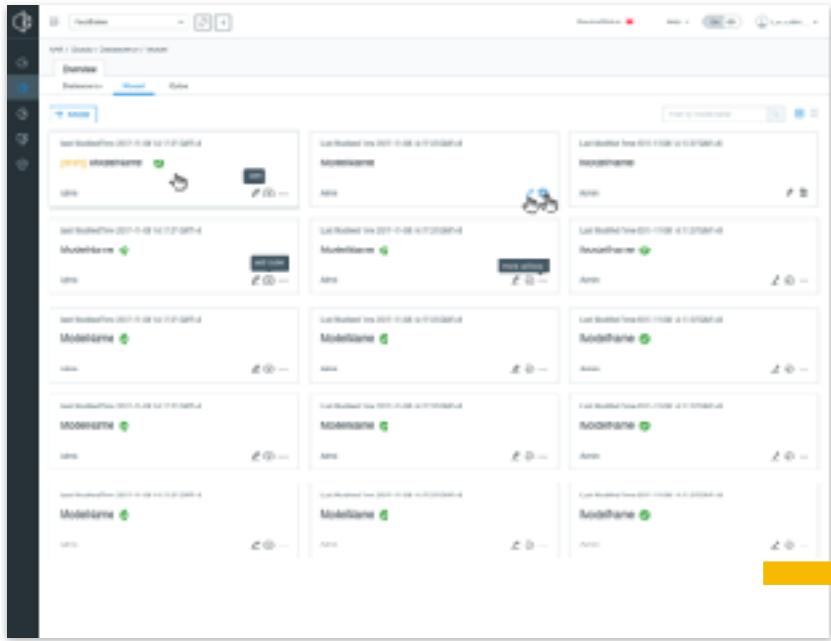
REDESIGN UI

KE first version was dark when I was joined Kyligence, but our clients start to complain it's hard to recognize wording on platform, so we started to plan 3 months to replace dark version to “white version” which is more clear and easier to read on old screen.

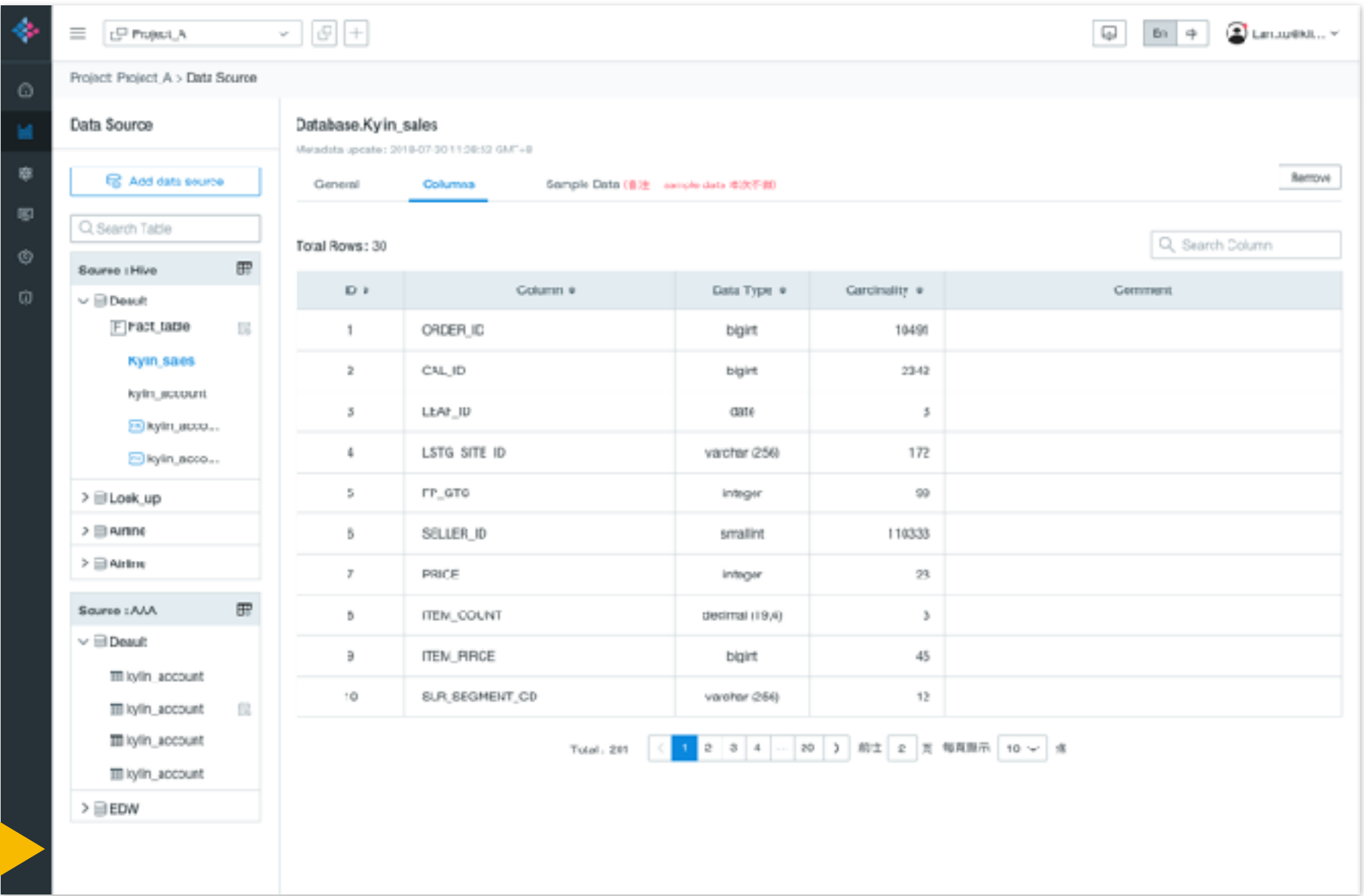
The Process of UI Redesign



P1 Dark Version



P2 Transition Period



P3 White Version applied to new guideline

KE IMPROVMENT

ENHANCED VISUALIZATION

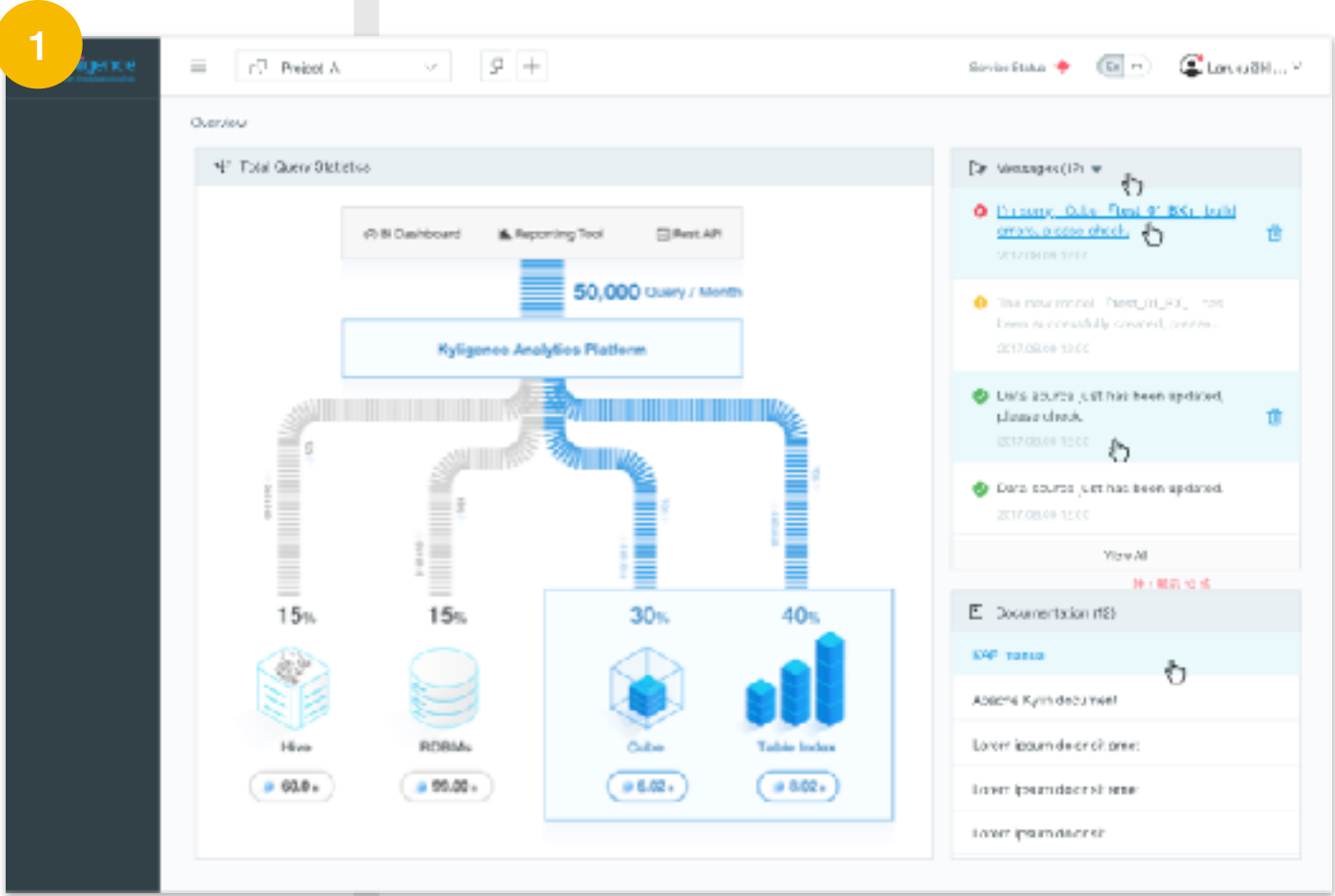
The original interface did not have an overview of the data to allow users to compare and analysis the data changing. Via the workshop revealed that users have a huge demand of info graphic issues, especially demonstrate the complex date in KE. So we designed many complicated diagrams and put these in our guideline.

Display Data (No.1,2)

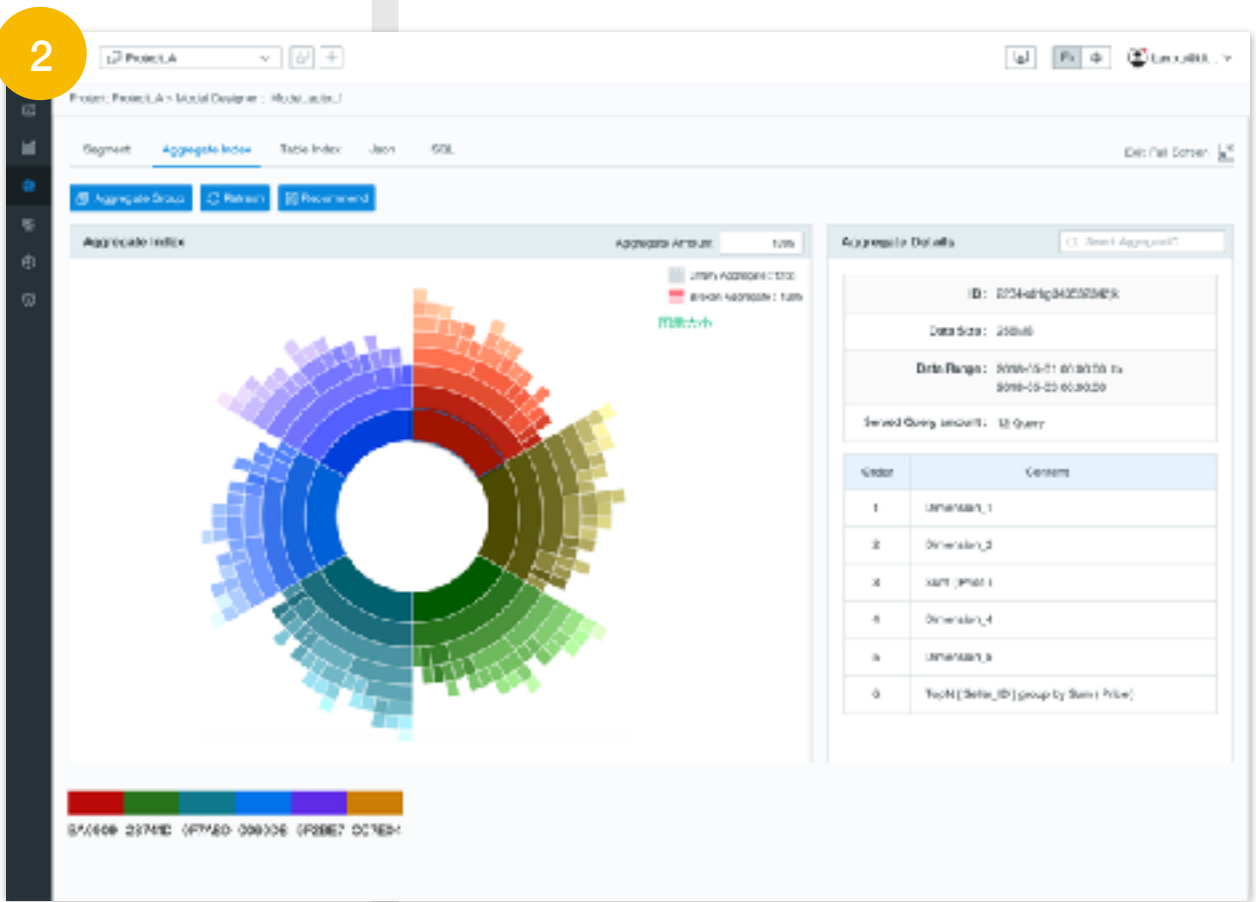
We can display the flow and health of data via the diagrams.

Monitor Hardware (No.3,4)

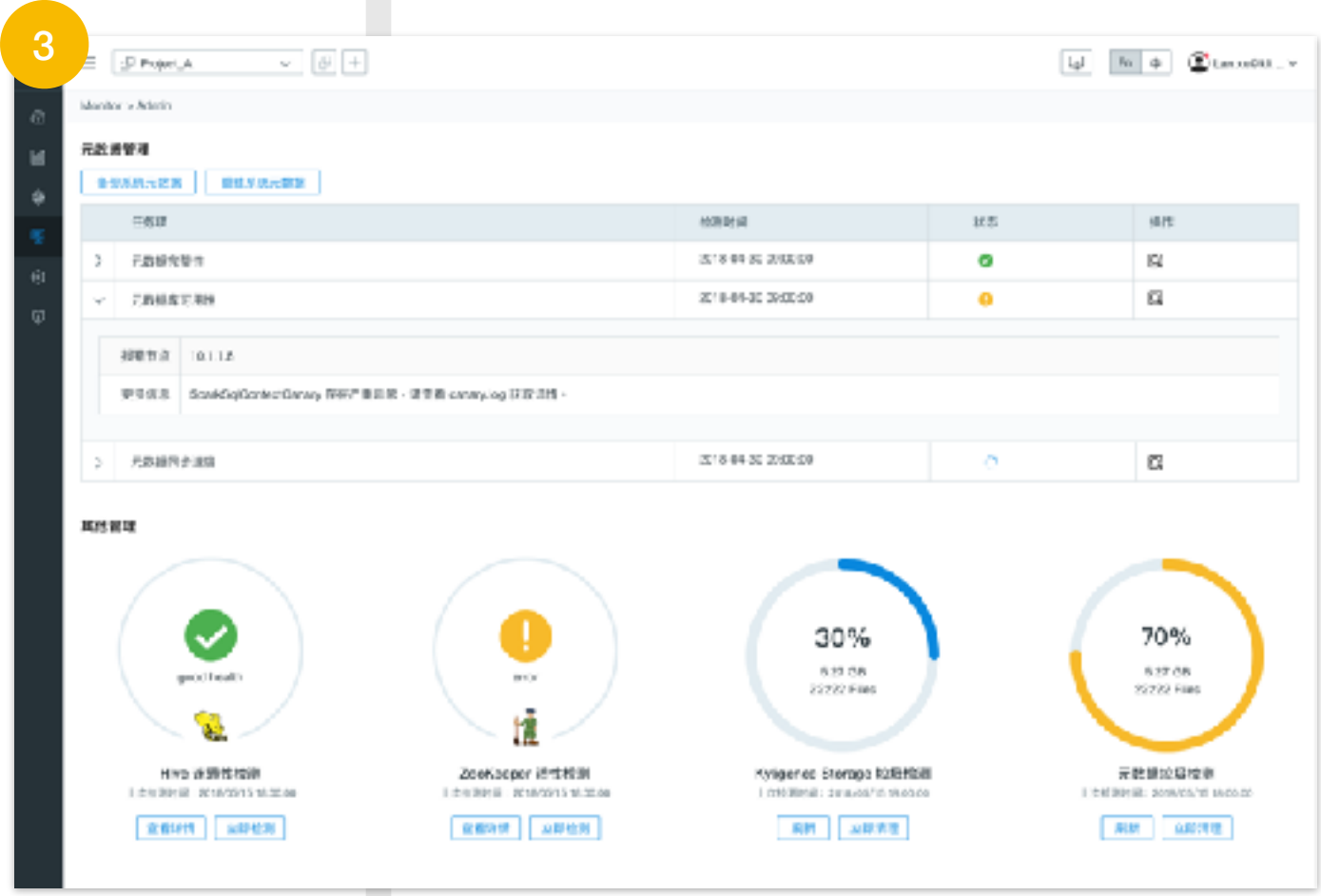
We can make use of diagrams to monitor cloud resources and status alert.



Monitor data flow on cloud



Monitor healthy of the dimension of aggregate group



Monitor hardware resources



Dashboard to demonstrate critical information

2017 Oct.

Joined
Kyligence

2017 Nov.

Started to design KC
& Designed guideline

2018 Feb.

KC beta GA &
Applied simplified
version of guideline

2018 May.

KC official GA

2018 Jun.

Started internal user
research & Replaced
dark version UI to KE
(3 weeks)

2018 Jul.

Applied new guideline
& Started improvement
project called Newton
(3 months)

2018 Oct.

Launched Newton
version & Applied
full version guideline
to KE

